CoFo AI — Ingestion Pipeline Specification

Architectural Blueprint for Multi-Format Content Transformation Engine

1. System Objective & Core Flow

CoFo AI ingests heterogeneous user inputs (raw prompts, messy URLs, YouTube links, PDFs, and job descriptions) and standardizes them into a unified **Canonical Intermediate Representation (IR)** before downstream format generation.

> **Pipeline Lifecycle Summary**
>
> User Payload (Multipart/JSON) → Gateway Triage & Regex Router → Small LLM Intent Classifier → Parallel Specialized Workers → Knowledge Normalizer → PostgreSQL Persistence.

2. Input Triage & Small LLM Router Specification

When a request hits the Express API endpoint, raw text is scanned via Regex to isolate embedded URLs while file attachments are staged. A small, cost-efficient LLM processes the triage metadata to output a deterministic routing plan.

Extraction Parameter	Detection Method	Target Worker
Web URLs	Regex URL matcher in prompt string	Firecrawl / Crawl4AI
YouTube Videos	Domain pattern match (youtube.com / youtu.be)	Transcript / Whisper API
Attached Documents	Multipart form-data MIME inspection	LlamaParse / Unstructured

3. Parallel Processing Workers

To prevent blocking socket connections during heavy ingestion tasks, background workers run concurrently using asynchronous execution queues. Each worker produces clean Markdown text blocks representing the raw source data.

Worker Module	Input Source	Output Artifact
Scraper Worker	Article or Job Posting URL	Clean Markdown (stripped ads/menus)
Media Worker	YouTube Link or MP3/MP4 File	Timestamped Transcript Text
Document Parser	PDF / Resume / Research Paper	Structured Sections, Tables & Text
Normalizer Worker	Aggregated Text Pools from all Workers	Canonical IR JSON Schema

4. Database Schema Design (PostgreSQL)

Persistence is structured across three relational tables to maintain strict audit trails of raw jobs, individual extracted source blocks, and final synthesized Canonical Intermediate Representations.

> **Core Table Structures**
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> 1. **`ingestion_jobs`** `(job_id PK, user_id, raw_prompt, target_output, tone, status)`
> 2. **`ingestion_sources`** `(source_id PK, job_id FK, source_type, source_identifier, content)`
> 3. **`canonical_records`** `(record_id PK, job_id FK, canonical_ir_json JSONB)`